

Linux System and Its Application



Student ID: 312023704026

Name: Adam Baha

Date: 05/04/2026

Lab3 File Contents operations

□ Objective

- Understand the functions of directory related commands: `cd`, `pwd`, `mkdir`, `rmdir`
- Master the usage of command `grep` and basic regular expression
- Be able to use regular expression locate specific string in files

□ Lab Environment

- Recommended: Ubuntu Desktop 24.04
- The other Linux or Unix-like environment

□ Instructions

- Login to **the account of your student id**. Use screenshot to show that you finish a specific task by paste it right below the tasks (**The screenshot should contain your student id**).
- Write answers to the questions related to each task right below the question.
- Once you finished, you should rename the current file name “Linux2026_lab3_STUID.docx” (**STUID should be changed to your own student id**) and then submit it to the Chaoxing system before 5th April.

□ Tasks

1. Download the file 'Lab3_resources.tar.gz' to your home directory and complete the tasks below.

- 1) Open terminal and verify your current directory using `pwd`.

Answer: `/home/312023704026`

- 2) In your home directory, use one-line command to create the following directory structure: `projects/linux/lab3/work/temp`.

Answer: `mkdir -p projects/linux/lab3/work/temp`

- 3) Enter the directory `temp`, then copy the file `Lab3_resources.tar.gz` from your home directory into this folder.

Answer: `cp ~/Downloads/Lab3_resources.tar.gz ~/projects/linux/lab3/work/temp/`

- 4) Go back to your home directory and extract `Lab3_resources.tar.gz` into the folder: `projects/linux/lab3/work/temp`.

Answer: `tar -xzf ~/Downloads/Lab3_resources.tar.gz -C ~/projects/linux/lab3/work/temp/`

```

312023704026@ubuntu: ~/projects/linux/lab3
312023704026@ubuntu:~$ mkdir -p projects/linux/lab3/work/temp
312023704026@ubuntu:~$ cp -r /Downloads/lab3_resources.tar.gz -r projects/linux/lab3/work/temp/
312023704026@ubuntu:~$ tar -xzf - /Downloads/lab3_resources.tar.gz -C -/projects/linux/lab3/work/temp/
note1.txt
note2.txt
employee.txt

```

- 5) Enter the directory *projects/linux*. Archive and compress the entire *lab3* folder into a file named *backup_lab3.tar.gz*, and display the contents of the archive without extracting it.

Answer: `tar -czvf backup_lab3.tar.gz lab3` and `tar -tvf backup_lab3.tar.gz`

- 6) Enter the directory *lab3/work/temp*. Merge *note1.txt* and *note2.txt* into a new file named *notes_all.txt*, and add the content of *notes_all.txt* to the end of *note2.txt*.

Answer: `cat note1.txt note2.txt > notes_all.txt` and `cat notes_all.txt >> note2.txt`

```

312023704026@ubuntu:~/projects/linux$ tar -czvf backup_lab3.tar.gz lab3
lab3/
lab3/work/
lab3/work/temp/
lab3/work/temp/employee.txt
lab3/work/temp/note2.txt
lab3/work/temp/note1.txt
lab3/work/temp/Lab3_resources.tar.gz
312023704026@ubuntu:~/projects/linux$ tar -tvf backup_lab3.tar.gz
drwxrwxr-x 312023704026/ubuntu 0 2026-04-04 14:04 lab3/
drwxrwxr-x 312023704026/ubuntu 0 2026-04-04 14:04 lab3/work/
drwxrwxr-x 312023704026/ubuntu 0 2026-04-04 14:55 lab3/work/temp/
-rw-rw-r-- 312023704026/ubuntu 886 2026-04-29 14:51 lab3/work/temp/employee.txt
-rw-rw-r-- 312023704026/ubuntu 228 2026-03-29 14:52 lab3/work/temp/note2.txt
-rw-rw-r-- 312023704026/ubuntu 224 2026-03-29 14:52 lab3/work/temp/note1.txt
-rw-rw-r-- 312023704026/ubuntu 905 2026-04-04 14:54 lab3/work/temp/Lab3_resources.tar.gz
312023704026@ubuntu:~/projects/linux$ cd lab3/work/temp
312023704026@ubuntu:~/projects/linux/lab3/work/temp$ cat note1.txt note2.txt > notes_all.txt
312023704026@ubuntu:~/projects/linux/lab3/work/temp$ cat notes_all.txt >> note2.txt

```

- 7) Display the inode numbers of all *.txt* files inside the *temp* folder.

Answer: `ls -li *.txt`

• Result: 9276 employee.txt, 9274 note1.txt, 9275 note2.txt, 13326 notes_all.txt

- 8) Move *employee.txt* to *lab3*, and change your current directory to *lab3*.

Answer: `mv employee.txt ../..` and `cd ../..`

- 9) Delete the directory *work* together with all its contents.

Answer: `rm -rf work`

```

312023704026@ubuntu:~/cs/linux/lab3/work/temp$ ls -li *.txt
9276 employee.txt 9274 note1.txt 9275 note2.txt 13326 notes_all.txt
312023704026@ubuntu:~/cs/linux/lab3/work/temp$ mv employee.txt ../..
312023704026@ubuntu:~/cs/linux/lab3/work/temp$ cd ../..
312023704026@ubuntu:~/cs/linux/lab3$ rm -rf work

```

2. Finish the tasks with `grep` command using the file 'employee.txt' (If

you finished task1, it should be in the directory ~/projects/linux/lab3).
Each line follows the format:

FirstName LastName|Department|City|Birthday|Salary

Example:

Alice Brown|HR|Seattle|04/15/1988|52000

1) Display all lines that start with letter A.

Answer: `grep "^A" employee.txt`

• Result: Alice Brown|HR|Seattle|04/15/1988|52000

2) Display all employees whose last name contains exactly 6 letters.

Answer: `grep "[a-zA-Z]\{6\}" employee.txt`

• Result: Cindy Miller, Emma Taylor, Frank Martin, Henry Walker, Jack Wilson, Kevin Harris, Oscar Garcia, Sophia Thomas.

```
312023704026@ubuntu:~/projects/linux/lab3$ grep "^A" employee.txt
Alice Brown|HR|Seattle|04/15/1988|52000
312023704026@ubuntu:~/projects/linux/lab3$ grep "[a-zA-Z]\{6\}" employee.txt
grep: Unmatched \{
312023704026@ubuntu:~/projects/linux/lab3$ grep "[a-zA-Z]\{6\}" employee.txt
Cindy Miller|Finance|Chicago|07/21/1985|73500
Emma Taylor|Marketing|New York|12/30/1987|51000
Frank Martin|HR|Dallas|09/09/1983|62000
Henry Walker|HR|San Jose|12/25/1989|101000
Jack Wilson|Finance|Boston|11/14/1984|101000
Kevin Harris|IT|New York|05/05/1986|72000
Oscar Garcia|IT|Boston|08/08/1982|109000
Sophia Thomas|Finance|Dallas|07/07/1985|77000
```

3) Display all employees whose: first name has 5 or 6 letters, and last name begins with M or T.

Answer: `grep -E "^[a-zA-Z]{5,6} [MT]" employee.txt`

4) Display all employees born in December.

Answer: `grep "|12/" employee.txt`

• Result: Brian Smith, Emma Taylor, Henry Walker, Linda Clark, Nancy Martinez, Robert Anderson, Tom Jackson.

```
312023704026@ubuntu:~/projects/linux/lab3$ grep -E "^[a-zA-Z]{5,6} [MT]" employee.tx
312023704026@ubuntu:~/projects/linux/lab3$ grep "|12/" employee.txt
Brian Smith|IT|New York|12/03/1990|68000
Emma Taylor|Marketing|New York|12/30/1987|51000
Henry Walker|IT|San Jose|12/25/1989|101000
Linda Clark|HR|Chicago|12/01/1993|53000
Nancy Martinez|Finance|Seattle|12/22/1990|88000
Robert Anderson|IT|San Jose|12/12/1991|84000
Tom Jackson|Marketing|Seattle|12/19/1989|56000
```

5) Display all employees working in New York or Chicago.

Answer: `grep -E "New York|Chicago" employee.txt`

- Result: Brian Smith, Cindy Miller, Emma Taylor, Grace Lee, Kevin Harris, Linda Clark, Paul Thompson, Queen Ramirez.

6) Display line numbers and content for all employees whose salary has 6 digits.

Answer: `grep -n "[0-9]\{6\}" employee.txt`

- Result:

- 8:Henry Walker|IT|San Jose|12/25/1989|101000

- 10:Jack Wilson|Finance|Boston|11/14/1984|101000

- 15:Oscar Garcia|IT|Boston|08/08/1982|109000

```
312023704026@ubuntu:~/projects/linux/lab3$ grep -E "New York|Chicago" employee.txt
Brian Smith|IT|New York|12/03/1990|68000
Cindy Miller|Finance|Chicago|07/21/1985|73500
Emma Taylor|Marketing|New York|12/30/1987|51000
Grace Lee|Finance|Chicago|01/11/1995|57000
Kevin Harris|IT|New York|05/05/1986|72000
Linda Clark|HR|Chicago|12/01/1993|53000
Paul Thompson|Marketing|Chicago|03/03/1994|45000
Queen Ramirez|HR|New York|10/10/1987|61000
312023704026@ubuntu:~/projects/linux/lab$ grep -n "[0-9]\{6\}" employee.txt
8:Henry Walker|IT|San Jose|12/25/1989|101000
10:Jack Wilson|Finance|Boston|11/14/1984|101000
15:Oscar Garcia|IT|Boston|08/08/1982|109000
312023704026@ubuntu:~/projects/linux/lab$
```